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# CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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## ABSTRACT

Adversarial training improves robustness but often sacrifices clean accuracy. We mitigate this trade-off by first learning a clean self-teacher from natural images and then preserving its knowledge during adversarial training. We propose Clean Feature Anchoring (CFA), which anchors the student’s adversarial representation to the teacher’s fixed clean representation while retaining its classifier, without a label objective. We connect this formulation to fixed clean-target regression and show that the feature anchor jointly controls clean teacher fidelity and local feature variation. We further introduce sensitivity-matched perturbation budgets to account for example-dependent responses to adversarial perturbations. On CIFAR-100 with ResNet-18, CFA achieves  $64.82 \pm 0.07\%$  clean and  $28.04 \pm 0.09\%$  AutoAttack accuracy, with consistent results across CIFAR-10, Tiny-ImageNet, and WideResNet-34-10.

## 1 INTRODUCTION

Adversarial training improves robustness but often reduces clean accuracy (Madry et al., 2017; Tsipras et al., 2018; Zhang et al., 2019). We ask whether robustness can be learned while retaining more of the representation that supports strong clean performance. Our approach separates these roles: we first train a clean self-teacher only on natural images, then adversarially train a student while preserving the teacher’s clean knowledge. The teacher is not expected to be robust; it provides the clean reference that the student seeks to retain.

This perspective differs from robust distillation methods that transfer predictive distributions and balance them against label supervision (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024). We instead use the clean teacher output as a fixed target throughout the student’s adversarial neighborhood. This leads to **Clean Feature Anchoring (CFA)**: the student’s adversarial feature is matched to the teacher’s clean feature while the inherited classifier remains fixed. The adversarial-training stage therefore uses no label objective.

Our analysis connects softened predictive supervision to direct regression on a fixed clean target and characterizes what the feature anchor controls. It also shows that teacher clean accuracy alone does not determine how much clean performance transfers to the robust student. Finally, because the same pixel radius can affect the anchor loss differently across examples, we redistribute a fixed batch budget of attack radii according to local anchor sensitivity. Thus the teacher specifies *what* should be preserved, while sensitivity matching determines *where* that constraint is enforced most strongly.

Empirically, CFA improves the clean-robust trade-off across CIFAR-10, CIFAR-100, Tiny-ImageNet, and WideResNet-34-10. On CIFAR-100 with ResNet-18, it achieves  $64.82 \pm 0.07\%$  clean and  $28.04 \pm 0.09\%$  AutoAttack accuracy across three runs. Our contributions are: (i) a clean self-teacher framework that preserves clean knowledge during label-free adversarial training; (ii) an analysis of fixed clean-target transfer, including a joint fidelity-variation bound and evidence that teacher accuracy alone does not characterize transferability; and (iii) sensitivity-matched per-

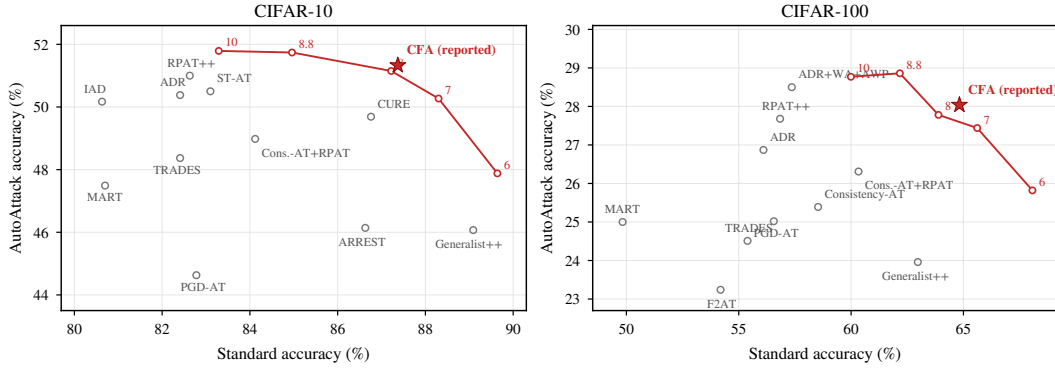


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at  $\epsilon = 8/255$ . Grey points are published results (Table 27); the CFA curve varies the training radius over 100-epoch runs, and the star denotes the reported 150-epoch result.

turbation budgets that improve clean accuracy without an observed loss of robustness in the final CIFAR-100 configuration.

## 2 RELATED WORK

**Adversarial training and its targets.** PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean-robust trade-off.

**Predictive targets in adversarial distillation.** ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. Our target instead comes from a natural teacher at the clean input and remains fixed during the student’s inner maximization.

**Guidance from naturally trained models.** BGAT is the closest direct-regression precedent: it matches adversarial logits to a fixed natural teacher’s clean logits, and LBGAT extends this to a jointly trained natural branch (Cui et al., 2021). Our logit control confirms that this fixed-clean-target principle is not specific to features. CFA uses the full clean representation with an inherited fixed classifier, and our analysis focuses on what a single anchor controls, which teacher checkpoint supplies it, and how its neighborhood is allocated. ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024).

**Sample-dependent perturbation budgets.** IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

## 3 ANALYSIS

We analyze how clean self-teacher knowledge should be transferred during adversarial training. We first connect predictive distillation to fixed clean-target regression, then characterize the feature anchor used by CFA, and finally examine how teacher clean performance relates to transfer.

### 3.1 FROM PREDICTIVE DISTILLATION TO A FIXED CLEAN TARGET

Let  $f = h \circ \Phi$ , with feature map  $\Phi$  and linear classifier  $h(z) = Wz + b$ . Subscripts  $t$  and  $s$  denote the frozen clean teacher and student. For clean  $x$  and  $x' \in B(x, \epsilon)$ , shared-temperature KD uses

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau), \quad (1)$$

and  $\tau^2 D_{\text{KL}}(q_t^\tau \| p_s^\tau)$ . For fixed logits, with  $\bar{f} = f - C^{-1} \mathbf{1} \mathbf{1}^\top f$ ,

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (2)$$

Thus high-temperature predictive distillation approaches centered-logit regression. CFA applies the same fixed-clean-target principle in feature space:

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (3)$$

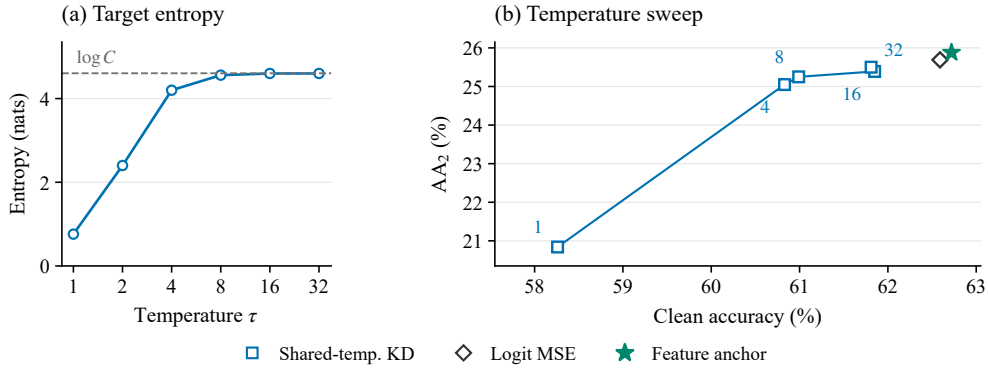


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Entropy of the clean teacher target as the shared temperature increases. (b) Clean accuracy and AA under shared-temperature KD, direct logit regression, and feature regression. Exact values are in Table 8.

Feature matching is a design choice rather than a consequence of Equation (2); empirically, direct logit and feature regression reach similar operating points (Figure 2 and table 11).

A key distinction is where the teacher is evaluated. A perturbed teacher feature can inherit the vulnerability of the natural teacher: replacing the fixed target  $\Phi_t(x)$  with  $\Phi_t(x_{adv})$  yields 76.11% clean accuracy but 0.00% AA, whereas the fixed clean target reaches 61.33/26.19 clean/AA under the same 50-epoch control (Table 6). This supports using the natural teacher as a clean reference rather than asking the student to reproduce its perturbed response.

### 3.2 WHAT THE ANCHOR CONTROLS

For  $B = B(x, \epsilon)$ , define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (4)$$

**Proposition 1** (Fixed-anchor decomposition). *For every  $x$  and  $\epsilon \geq 0$ ,  $L \leq F + O \leq 3L$ .*

The result follows from the triangle inequality; the proof and scope are in Appendix A. Hence  $L \leq \delta$  implies  $F \leq \delta$  and  $O \leq 2\delta$ : one fixed anchor controls both clean teacher fidelity and local feature variation, without requiring the teacher itself to be robust. The same argument applies to any norm-based fixed target, including logits, and does not establish that feature coordinates are inherently superior.

### 3.3 TEACHER CLEAN PERFORMANCE AND TRANSFERABILITY

Higher teacher clean accuracy need not produce a student with higher clean accuracy after adversarial training. Along one CIFAR-100 teacher trajectory, teacher clean accuracy rises from 75.81% to 78.32%, while student clean accuracy falls from 64.28% to 62.24% and AA rises from 24.38% to 25.80% (??). An independent trajectory shows the same overall pattern (Table 22). Moreover, label-smoothed and mixup teachers differing by only 0.14 points in clean accuracy produce students differing by 4.18 clean points, with nearly identical AA (??). Teacher clean accuracy is therefore an incomplete descriptor of transferability. We fix the 200-epoch teacher as a reference for the remaining experiments.

## 4 METHOD

The analysis motivates two components: a fixed clean representation target and a sample-dependent perturbation budget.

## 4.1 CLEAN FEATURE ANCHORING

Let  $\Phi_t, h_t$  denote the frozen clean self-teacher. The student inherits  $\Phi_s \leftarrow \Phi_t$  and  $h_s = h_t$ , after which only  $\Phi_s$  is optimized. For each clean example,

$$x_{\text{adv}} = \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \quad \mathcal{L}_{\text{CFA}} = \mathbb{E}_x \|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2. \quad (5)$$

We approximate the inner maximization with randomly initialized PGD. No label loss is used in this stage, and inference uses  $h_t \circ \Phi_s$ . The full procedure is given in Algorithm 1.

## 4.2 SENSITIVITY-MATCHED $\epsilon$

A uniform pixel radius can induce different changes in the anchor loss across examples. For  $\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2$ , first-order movement in an  $\ell_\infty$  ball satisfies

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_{x'} \ell_{\text{anc}}, \delta \rangle = e \|\nabla_{x'} \ell_{\text{anc}}\|_1. \quad (6)$$

Motivated by this relation, let  $g_i > 0$  be a local sensitivity score. We allocate

$$\epsilon_i = N \epsilon \frac{g_i^{-p}}{\sum_{j=1}^N g_j^{-p}}, \quad (7)$$

so  $\sum_i \epsilon_i = N \epsilon$ . Thus the method redistributes, rather than increases, the batch perturbation budget;  $p = 0$  recovers a uniform radius. Implementation details, clipping, and the sensitivity-norm control are reported in Appendix F.

# 5 EXPERIMENTS

We evaluate whether CFA improves the clean-robust trade-off, generalizes beyond the default setting, and whether its two defining components account for the gain.

## 5.1 EXPERIMENTAL SETTINGS

We evaluate CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009), and Tiny-ImageNet-200 (Le & Yang, 2015), using ResNet-18 by default and WideResNet-34-10 for model-scale transfer. Unless stated otherwise, the clean self-teacher is trained on natural images for 200 epochs and frozen; the student inherits its backbone and fixed classifier. Main CIFAR runs use 150 epochs, AdamW, a one-cycle schedule, PGD-10,  $\epsilon_{\text{train}} = 8/255$ , sensitivity matching ( $p = 1$ ), WA, and AWP. We report clean and AutoAttack (AA) accuracy at 8/255, their arithmetic mean (Avg), and harmonic mean (NRR). Full settings and attack diagnostics are in Appendices G and J.

## 5.2 MAIN RESULTS

On CIFAR-100, CFA reaches 64.82% clean and 28.04% AA (NRR 39.15); relative to ADR, the closest baseline in NRR, this is +4.89 clean, +0.84 AA, and +1.73 NRR points. On CIFAR-10, CFA reaches 87.37/51.33 clean/AA and an NRR of 64.67, exceeding the closest baseline NRR by 1.85 points (Table 1).

## 5.3 GENERALIZATION

CFA also transfers across data and model scale. On Tiny-ImageNet it reaches 55.16/20.54 clean/AA, and on CIFAR-10 with WideResNet-34-10 it reaches 88.67/55.29. On CIFAR-100 with the same architecture, CFA obtains 65.04/30.14, compared with ADR at 64.50/29.04 (Table 19).

## 5.4 ABLATION AND COMPONENT ANALYSIS

**Clean feature anchoring.** Without WA, AWP, or sensitivity matching, the anchor exceeds label cross-entropy by 6.69 clean and 5.91 AA points under the same recipe; it remains ahead when cross-entropy uses its own optimizer and schedule (Table 3). Together with the clean-versus-perturbed teacher-target control in Section 3.1, this isolates the fixed clean anchor from the surrounding training stack.

Table 1: ResNet-18 results under  $\ell_\infty$  perturbations with  $\epsilon = 8/255$ . All values are measured in our framework.  $T$  denotes the teacher assumed by each method: none (–), a natural model (nat.), or the method’s own weight average (self). The CIFAR-100 CFA row reports the mean of three student seeds; all other rows are individual runs.

| Method            | T.   | CIFAR-10     |              |              |              | CIFAR-100    |              |              |              |
|-------------------|------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                   |      | Clean        | AA           | Avg          | NRR          | Clean        | AA           | Avg          | NRR          |
| PGD-AT            | –    | 84.54        | 43.57        | 64.06        | 57.50        | 57.36        | 20.34        | 38.85        | 30.03        |
| TRADES            | –    | 81.71        | 49.16        | 65.44        | 61.39        | 55.28        | 23.55        | 39.42        | 33.03        |
| MART              | –    | 80.41        | 45.56        | 62.99        | 58.16        | 53.86        | 22.72        | 38.29        | 31.96        |
| ARD               | nat. | 84.58        | 43.75        | 64.16        | 57.67        | 57.61        | 20.24        | 38.93        | 29.96        |
| RSLAD             | nat. | 85.15        | 42.91        | 64.03        | 57.06        | 59.68        | 21.30        | 40.49        | 31.40        |
| AdaAD             | nat. | 85.39        | 46.22        | 65.81        | 59.98        | 59.79        | 23.19        | 41.49        | 33.42        |
| AdaAD + IGDM      | nat. | 85.01        | 45.50        | 65.26        | 59.27        | 48.25        | 19.48        | 33.87        | 27.75        |
| LBGAT             | nat. | –            | –            | –            | –            | 56.46        | 26.02        | 41.24        | 35.62        |
| ARREST            | nat. | 85.06        | 46.42        | 65.74        | 60.06        | <b>66.97</b> | 22.78        | 44.88        | 34.00        |
| HAT               | nat. | 84.59        | 48.81        | 66.70        | 61.90        | 60.67        | 24.45        | 42.56        | 34.85        |
| ADR               | self | 83.36        | 49.72        | 66.54        | 62.29        | 59.93        | 27.20        | 43.57        | 37.42        |
| RPAT++            | –    | 82.41        | 50.75        | 66.58        | 62.82        | 55.93        | 27.36        | 41.64        | 36.74        |
| <b>CFA (ours)</b> | nat. | <b>87.37</b> | <b>51.33</b> | <b>69.35</b> | <b>64.67</b> | <b>64.82</b> | <b>28.04</b> | <b>46.43</b> | <b>39.15</b> |

Table 2: Results on Tiny-ImageNet-200 with ResNet-18 and CIFAR-10 with WideResNet-34-10, measured in our framework.

| Method            | T.   | Tiny-ImageNet, ResNet-18 |              |              |              | CIFAR-10, WRN-34-10 |              |              |              |
|-------------------|------|--------------------------|--------------|--------------|--------------|---------------------|--------------|--------------|--------------|
|                   |      | Clean                    | AA           | Avg          | NRR          | Clean               | AA           | Avg          | NRR          |
| PGD-AT            | –    | 49.72                    | 14.45        | 32.09        | 22.39        | 86.00               | 45.71        | 65.86        | 59.69        |
| TRADES            | –    | 46.98                    | 15.95        | 31.46        | 23.83        | 85.13               | 48.99        | 67.06        | 62.19        |
| MART              | –    | 46.01                    | 16.90        | 31.46        | 24.73        | 85.29               | 47.06        | 66.18        | 60.65        |
| ARD               | nat. | 49.74                    | 14.48        | 32.11        | 22.43        | 86.35               | 45.52        | 65.94        | 59.61        |
| RSLAD             | nat. | 51.26                    | 17.02        | 34.14        | 25.56        | 86.41               | 45.80        | 66.10        | 59.87        |
| AdaAD             | nat. | 51.59                    | 16.79        | 34.19        | 25.33        | 86.80               | 46.84        | 66.82        | 60.85        |
| LBGAT             | nat. | 51.04                    | 18.47        | 34.75        | 27.13        | 81.56               | 51.79        | 66.68        | 63.35        |
| ARREST            | nat. | <b>59.38</b>             | 16.06        | 37.72        | 25.28        | 87.54               | 48.83        | 68.19        | 62.69        |
| HAT               | nat. | 52.57                    | 17.28        | 34.93        | 26.01        | 87.15               | 53.21        | 70.18        | 66.07        |
| ADR               | self | 47.66                    | <b>20.85</b> | 34.25        | 29.01        | 88.64               | 52.34        | 70.49        | 65.82        |
| <b>CFA (ours)</b> | nat. | 55.16                    | 20.54        | <b>37.85</b> | <b>29.93</b> | <b>88.67</b>        | <b>55.29</b> | <b>71.98</b> | <b>68.11</b> |

Table 3: Objective control on CIFAR-100 with ResNet-18, 100 epochs, no weight averaging or AWP,  $p = 0$ , and  $\epsilon_{\text{train}} = 8.8/255$ .

| Objective                        | Clean        | AA           | NRR          |
|----------------------------------|--------------|--------------|--------------|
| Anchor                           | <b>61.29</b> | <b>25.36</b> | <b>35.88</b> |
| Label cross-entropy, same recipe | 54.60        | 19.45        | 28.68        |
| Label cross-entropy, own recipe  | 56.66        | 20.91        | 30.55        |

Table 4: Clean-target regression in the final CIFAR-100 recipe (ResNet-18, 150 epochs,  $\epsilon_{\text{train}} = 8/255$ ). Feature rows report mean  $\pm$  sample standard deviation over three student seeds and differ only in  $p$ ; logit regression is one run at  $p = 1$ .

| Target and radius                         | Clean                              | AA                                 | NRR                                |
|-------------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| Features, uniform ( $p = 0$ )             | $62.63 \pm 0.22$                   | $28.01 \pm 0.20$                   | $38.71 \pm 0.21$                   |
| Features, sensitivity-matched ( $p = 1$ ) | <b><math>64.82 \pm 0.07</math></b> | <b><math>28.04 \pm 0.09</math></b> | <b><math>39.15 \pm 0.09</math></b> |
| Logits, sensitivity-matched ( $p = 1$ )   | 64.49                              | 27.85                              | 38.90                              |

**Sensitivity-matched radius allocation.** Across three paired runs, sensitivity matching improves clean accuracy by  $2.19 \pm 0.20$  points while AA changes by only  $0.03 \pm 0.23$  points. The clean gain appears in every seed, whereas the AA difference is within run-to-run variation. Direct logit regression remains close to feature regression, consistent with the fixed-target interpretation.

Table 5: Radius-allocation signals under the same mean radius on CIFAR-100 with ResNet-18, trained for 100 epochs with WA and AWP.  $\text{sd}(w)$  measures multiplier dispersion.

| Allocation                         | $\text{sd}(w)$ | Clean        | AA           | NRR          |
|------------------------------------|----------------|--------------|--------------|--------------|
| Uniform                            | 0              | 60.42        | 28.42        | 38.66        |
| Sensitivity                        | 0.32           | <b>62.17</b> | <b>28.86</b> | <b>39.42</b> |
| Same multipliers, difficulty order | 0.32           | 60.90        | 28.06        | 38.42        |
| Difficulty magnitude               | 0.39           | 61.27        | 28.12        | 38.55        |
| Logit margin                       | 0.42           | 61.07        | 27.79        | 38.20        |

**Does the allocation signal matter?** Reassigning the same sensitivity-derived radius multipliers by sample difficulty reduces AA by 0.80 points and NRR by 1.00 point, despite preserving the radius distribution. Thus heterogeneous radii alone do not explain the gain. Additional recipe, radius, and WA/AWP controls are reported in Tables 10, 15, 16 and 24.

## 6 CONCLUSION

CFA addresses the clean-robust trade-off by learning clean knowledge first and preserving it during adversarial training. A fixed clean target provides a simple alternative to softened predictive supervision, while the feature realization jointly controls teacher fidelity and local feature variation. Sensitivity-matched radii further improve retained clean accuracy without an observed AA loss in the final CIFAR-100 setting. The results across datasets, architectures, and controls support clean self-teaching as a practical route to robust training without an adversarial teacher or label objective during the robust stage.

## A PROOF AND SCOPE OF THE FIXED-ANCHOR DECOMPOSITION

*Proof of Proposition 1.* For any  $x' \in B(x, \epsilon)$ , the triangle inequality gives

$$\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + \|\Phi_s(x) - \Phi_t(x)\|_2 \leq O + F,$$

so  $L \leq F + O$ . Since  $x \in B(x, \epsilon)$ ,  $F \leq L$ . For any  $x', x'' \in B(x, \epsilon)$ ,

$$\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq \|\Phi_s(x') - \Phi_t(x)\|_2 + \|\Phi_s(x'') - \Phi_t(x)\|_2 \leq 2L,$$

so  $O \leq 2L$  and therefore  $F + O \leq 3L$ .  $\square$

The result concerns exact maxima over the perturbation set; finite-step PGD does not certify them. It applies to any fixed target in a normed space, including logits, and does not imply that features are superior to logits or that the bound alone guarantees robust classification.

## B ALGORITHM

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**Algorithm 1** Clean Feature Anchoring (CFA)

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1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi})$ ;  $w_i \leftarrow Nv_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i$ ;  $a_i \leftarrow aw_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

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## C TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

### C.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

The setting is the two-feature model of Tsipras et al. (2018), which is where the objection is usually posed. One feature  $x_1$  agrees with the label with probability  $p = 0.95$  and cannot be attacked. A further  $d$  features  $z_j \sim \mathcal{N}(\eta y, 1)$  are individually almost uninformative, are jointly close to perfectly informative through their mean  $m(z)$ , and can all be moved by an  $\ell_\infty$  adversary. A naturally trained network therefore reads  $m(z)$ : near-perfect clean accuracy, no robustness whatever. That is our teacher’s profile, so  $\Phi_t = m(z)$ .

Write the student’s feature as  $\Phi_s = ax_1 + bm(z)$ . Then  $b$  is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave  $b$  large.

**Proposition 2.** *With the inner maximum exact,  $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$  with  $R = ax_1 + (b - 1)m$ . Its one-sided slopes at  $b = 0$  differ by  $4\epsilon \mathbb{E}|R(a, 0)|$ ; above a threshold  $\epsilon_0$ , every minimizer has  $b^* = 0$ .*

*Proof.* The adversary reaches  $\Phi_s$  only through  $m$ , and only through the mean perturbation  $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$ , which ranges over  $[-\epsilon, \epsilon]$  as  $\|\delta\|_\infty \leq \epsilon$ . The inner objective is therefore  $\max_{|\bar{\delta}| \leq \epsilon} (R +$

$b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$ , since a scalar quadratic is maximized at an endpoint and the sign of  $\bar{\delta}$  is free. Taking expectations and expanding the square gives the display.

Fix  $a$  and differentiate in  $b$ . The term  $\mathbb{E}[R^2]$  and  $\epsilon^2 b^2$  are smooth, while  $2\epsilon|b| \mathbb{E}|R|$  contributes  $\pm 2\epsilon \mathbb{E}|R|$  as  $b \rightarrow 0^\pm$ . The one-sided derivatives at  $b = 0$  are thus  $2\mathbb{E}[R(a, 0) m] \pm 2\epsilon \mathbb{E}|R(a, 0)|$ , so  $b = 0$  is a minimum along  $b$  exactly when

$$|\mathbb{E}[R(a, 0) m]| \leq \epsilon \mathbb{E}|R(a, 0)|. \quad (8)$$

At  $b = 0$  the objective in  $a$  is  $\mathbb{E}(ax_1 - m)^2$ . Writing  $q = 2p - 1$  for the mean label-alignment of  $x_1$ , it is minimized at  $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$ , and there  $\mathbb{E}[R(a^*, 0) m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$  with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (9)$$

Equation (8) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}|q\eta x_1 - m|}, \quad (10)$$

which is the stated threshold. For fixed  $x_1, m$  and  $\bar{\delta} \in [-\epsilon, \epsilon]$ ,  $(ax_1 + (b - 1)m + b\bar{\delta})^2$  is convex in  $(a, b)$ . Its maximum over  $\bar{\delta}$  and then its expectation are convex, so  $J$  is jointly convex. At  $(a^*, 0)$  the  $a$ -derivative vanishes, and Equation (8) puts zero in the  $b$ -subgradient; hence this point is a global minimizer. If  $\epsilon > \epsilon_0$ , the directional derivative from  $(a^*, 0)$  is strictly positive in every direction with nonzero  $b$ . Along  $b = 0$ ,  $\mathbb{E}(ax_1 - m)^2$  has a unique minimizer. Thus every global minimizer has  $b = 0$  above the threshold.  $\square$

**Why the coefficient is deleted rather than shrunk.** The adversary contributes  $2\epsilon|b| \mathbb{E}|R|$  to the loss — linear in  $|b|$ , not quadratic in  $b$ . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on  $|b|$  is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At  $d = 512$  the model's constants give  $c = 7.89 \times 10^{-3}$  and  $\mathbb{E}|q\eta x_1 - m| = 0.0529$ , hence  $\epsilon_0 = 0.843 \eta$  — inside the bracket  $(0.5\eta, 1.0\eta)$  that the Monte-Carlo minimization returns, and far below the  $2\eta$  at which the model is usually stated.

The model assumes  $x_1$  is unperturbable and  $z$  fully perturbable, so some suppression of  $z$  is built in. The proposition establishes the exact threshold behavior within this restricted model; it does not describe the geometry of real image classifiers.

That the exactness is the kink's doing, and not a general consequence of adding an adversary, is visible from dropping it. With the  $|b|$  term removed,  $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$ ; minimizing out  $a$  and writing  $u = 1 - b$  collapses it to  $u^2 c + \epsilon^2 (1 - u)^2$ , so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (11)$$

which is strictly positive at every radius. Against the exact minimizer ( $d = 512$ , Monte-Carlo,  $n = 2 \times 10^5$ ):

| $\epsilon$ | $b^*$ (exact) | $b_{\text{quad}}^*$ (kink dropped) |
|------------|---------------|------------------------------------|
| $0.5\eta$  | 0.510         | 0.502                              |
| $1.0\eta$  | <b>0.000</b>  | 0.202                              |
| $2.0\eta$  | <b>0.000</b>  | 0.060                              |
| $4.0\eta$  | <b>0.000</b>  | 0.016                              |

The two columns agree below  $\epsilon_0$  and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every  $\epsilon$  in  $J$  multiplies  $b$ , at  $b^* = 0$  the objective reduces to  $\mathbb{E}(ax_1 - m)^2$  and no longer depends on the radius. This saturation is a property of the two-feature model, not a prediction of the empirical radius sweep in Table 15.

The two-feature model cannot explain the measured clean–robust trade-off: it has only a robust and a fragile direction. It shows the narrower point that a robust student need not copy the fragile route by which its natural teacher computes the clean target.

## C.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing  $\gamma_t(x)$  for the teacher’s margin at  $x$  and  $D_y$  for the relevant decision-boundary constant, Cauchy–Schwarz gives that  $\gamma_t(x) > D_y \cdot L(x)$  implies the student is robustly correct at  $x$ . The certificate is vacuous at the scales we train at, but the ratio  $DL/\gamma_t$  tracks both axes across the teacher ladder of ???. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

## D SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

### D.1 TEACHER EVALUATION ON PERTURBED INPUTS

The comparison in Section 3.1 evaluates the teacher at clean inputs. Here we examine why its responses to perturbed inputs may be unsuitable as supervisory targets. Under  $\epsilon = 8/255$ , the natural teacher’s feature rotates by  $63.8^\circ$  and its norm grows by a factor of 2.45. Its representation can therefore change substantially within the perturbation ball.

As a control, replacing the clean target  $\Phi_t(x)$  with  $\Phi_t(x')$  makes an exact copy of the teacher,  $\Phi_s = \Phi_t$ , a zero-loss solution even when the teacher is sensitive to perturbations. This variant yields 0.00% AutoAttack accuracy (Table 6). With a fixed clean target, copying the teacher instead incurs a loss wherever its features change within the ball.

Table 6: Where the teacher is evaluated. CIFAR-100, ResNet-18, 50 epochs; the two rows differ only in the teacher input used by the loss and inner maximization.

| Teacher target           | Clean        | AA           | NRR          |
|--------------------------|--------------|--------------|--------------|
| $\Phi_t(x)$              | 61.33        | <b>26.19</b> | <b>36.71</b> |
| $\Phi_t(x_{\text{adv}})$ | <b>76.11</b> | 0.00         | 0.00         |

IGDM (Lee et al., 2025) uses the finite difference  $f_t(x + \delta) - f_t(x - \delta)$  to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 12). For the natural teacher, the finite difference has norm 14.14, compared with  $\|f_t(x)\| = 18.36$ , and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 17). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.1, although differences among training recipes prevent it from isolating the target alone.

### D.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

We measure feature geometry on the final 150-epoch CFA model and a label-trained student with the same natural initialization, weight averaging, and AWP. Both models are evaluated on 5,000 test images under the same 10-step true-label PGD attack at  $8/255$ ; all angular statistics use unit-normalized features.

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. Under the matched true-label PGD protocol, the anchor and label-trained students move by 0.312 and 0.281 of their clean feature norms. These displacements are not the worst-case diameter  $O$  and provide no evidence of smaller relative displacement under anchoring. The reproducible difference is instead within-class concentration: CFA has a  $5.3\times$  smaller clean  $S_w/S_b$  and a  $2.85\times$  smaller ratio under attack (Table 7). Its class centroids are not farther apart

than those learned with labels, so the result supports tighter clusters rather than greater inter-class separation.

Table 7: Feature geometry of the final models under the same true-label PGD attack. Angles are measured on unit-normalized features. *Own* is the sample-to-own-centroid angle; *gap* is the mean angle from each class centroid to its nearest other-class centroid.

|          | Clean        |              |              | Adversarial  |              |              |
|----------|--------------|--------------|--------------|--------------|--------------|--------------|
|          | $S_w/S_b$    | Own          | Gap          | $S_w/S_b$    | Own          | Gap          |
| CFA      | <b>0.709</b> | <b>26.3°</b> | 29.2°        | <b>1.969</b> | <b>33.5°</b> | 19.2°        |
| Label CE | 3.744        | 52.2°        | <b>33.5°</b> | 5.611        | 55.0°        | <b>28.2°</b> |

The classification benefit is evaluated in Section 5.4. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 3, Table 23). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, CFA has slightly larger relative feature displacement and smaller centroid gaps, but substantially tighter class-conditional features. This association does not establish concentration as the cause of the robustness gain.

The checkpoint ladder in ?? further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9°, whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9°. The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8°—while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

The target comparison uses the 50-epoch base regime at  $\epsilon = 8/255$ , without weight averaging or AWP.

Table 8: Numerical values for the target comparison in Figure 2.  $H(q_t^\tau)$  is measured in nats, and NRR is the harmonic mean of clean and AA accuracy.

| Target                            | $\tau$ | $H(q_t^\tau)$ | Clean        | AA           | NRR          |
|-----------------------------------|--------|---------------|--------------|--------------|--------------|
| <i>Teacher-only temperature</i>   |        |               |              |              |              |
| KL                                | 1      | 0.76          | 58.26        | 20.84        | 30.70        |
| KL                                | 2      | 2.40          | 59.33        | 22.79        | 32.93        |
| KL                                | 4      | 4.20          | 59.39        | 24.48        | 34.67        |
| KL                                | 8      | 4.56          | 59.47        | 25.19        | 35.39        |
| KL                                | 16     | 4.60          | 57.78        | 24.00        | 33.91        |
| <i>Shared-temperature KD</i>      |        |               |              |              |              |
| KL                                | 4      | 4.20          | 60.83        | 25.05        | 35.49        |
| KL                                | 8      | 4.56          | 60.99        | 25.25        | 35.71        |
| KL                                | 16     | 4.60          | 61.85        | 25.39        | 36.00        |
| KL                                | 32     | 4.60          | 61.81        | 25.50        | 36.11        |
| Logit regression, fixed head      | —      | —             | 62.59        | 25.69        | 36.43        |
| <b>Feature anchor, fixed head</b> | —      | —             | <b>62.72</b> | <b>25.88</b> | <b>36.64</b> |

Table 9: Numerical values for the checkpoint trends in ??.

| Checkpoint | Teacher      |              | Student      |              |              |
|------------|--------------|--------------|--------------|--------------|--------------|
|            | Clean        | $S_w/S_b$    | Clean        | $S_w/S_b$    | AA           |
| 50 epochs  | 75.81        | 1.100        | <b>64.28</b> | 0.955        | 24.38        |
| 100 epochs | 76.62        | 0.982        | 63.65        | 0.900        | 25.20        |
| 150 epochs | 77.52        | 0.887        | 62.93        | 0.875        | 25.78        |
| 200 epochs | 77.66        | 0.808        | 62.72        | 0.850        | <b>25.88</b> |
| 300 epochs | <b>78.32</b> | <b>0.712</b> | 62.24        | <b>0.829</b> | 25.80        |

## E ADDITIONAL RESULTS

**Baseline training-stack control.** To test whether CFA’s final performance can be explained by teacher initialization, weight averaging, and AWP alone, we add the same components to the baseline recipes. Most baselines improve, but none reaches CFA’s NRR.

Table 10: Adding teacher initialization, weight averaging, and AWP to baseline recipes on CIFAR-100 (ResNet-18).

| Method            | Own recipe |       |       | + teacher init + WA + AWP |              |              |
|-------------------|------------|-------|-------|---------------------------|--------------|--------------|
|                   | Clean      | AA    | NRR   | Clean                     | AA           | NRR          |
| PGD-AT            | 57.36      | 20.34 | 30.03 | 59.85                     | 25.77        | 36.03        |
| TRADES            | 55.28      | 23.55 | 33.03 | 56.16                     | 24.56        | 34.17        |
| MART              | 53.86      | 22.72 | 31.96 | 50.97                     | 25.13        | 33.66        |
| ARD               | 57.61      | 20.24 | 29.96 | 60.19                     | 26.10        | 36.41        |
| RSLAD             | 59.68      | 21.30 | 31.40 | 63.44                     | 25.55        | 36.43        |
| AdaAD             | 59.79      | 23.19 | 33.42 | 57.21                     | 27.10        | 36.78        |
| AdaAD + IGDM      | 48.25      | 19.48 | 27.75 | 54.24                     | 25.56        | 34.75        |
| Consistency-AT    | 57.53      | 20.01 | 29.69 | 60.74                     | 25.38        | 35.80        |
| LBGAT             | 56.46      | 26.02 | 35.62 | 53.91                     | 25.16        | 34.31        |
| ADR               | 59.93      | 27.20 | 37.42 | 59.84                     | 27.84        | 38.00        |
| HAT               | 60.67      | 24.45 | 34.85 | 57.63                     | 24.74        | 34.62        |
| ARREST            | 66.44      | 22.99 | 34.16 | 64.95                     | 23.02        | 33.99        |
| <b>CFA (ours)</b> | —          |       |       | <b>64.82</b>              | <b>28.04</b> | <b>39.15</b> |

The following tables provide additional results supporting the main experiments and supplementary analysis.

**Regression under weight averaging and AWP.** Table 11 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 11: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform  $\epsilon_{\text{train}} = 8.8/255$ . Evaluation uses AA at  $8/255$ . Both losses sum squared errors across output coordinates and average over the batch.

| Target         | Clean | AA    |
|----------------|-------|-------|
| Clean logits   | 60.36 | 28.02 |
| Clean features | 60.42 | 28.42 |

**Normalization check.** Because Section 3.3 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared  $\ell_2$  as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 12: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an  $\epsilon = 8/255$  ball, computed as in IGDM’s Sec. 3.1.

| Network                                                   | Remainder proportion  |
|-----------------------------------------------------------|-----------------------|
| IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published) | 0.012 / 0.012 / 0.016 |
| PGD-AT ResNet-18 (ours)                                   | 0.0111                |
| <b>Natural teacher (ours)</b>                             | <b>0.4763</b>         |
| Anchored student, $p = 0$                                 | 0.0029                |
| Anchored student, reported recipe                         | 0.0202                |

Table 13: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP,  $\epsilon = 8/255$ ), so that no stack component can absorb the difference.

|                                       | Clean        | PGD-20 | CW           | AA           | NRR          |
|---------------------------------------|--------------|--------|--------------|--------------|--------------|
| Logit target only, $\tau = 1$         | 58.26        | 22.62  | 22.79        | 20.84        | 30.70        |
| Logit target only, $\tau = 4$         | 59.39        | 30.30  | 26.84        | 24.48        | 34.67        |
| Logit target only, $\tau = 16$        | 57.78        | 31.57  | 26.11        | 24.00        | 33.91        |
| Feature anchor + head KD, $\tau = 1$  | 62.34        | 26.34  | 26.11        | 23.93        | 34.58        |
| Feature anchor + head KD, $\tau = 16$ | 62.61        | 32.47  | 27.32        | 25.61        | 36.35        |
| <b>Feature anchor, head untouched</b> | <b>62.72</b> | 28.69  | <b>27.80</b> | <b>25.88</b> | <b>36.64</b> |
| head refit, adversarial CE            | 62.77        | 29.93  | 27.66        | 25.65        | 36.42        |
| head refit, adv. CE + smoothing 0.1   | 62.57        | 30.43  | 27.55        | 25.66        | 36.39        |
| head refit, clean CE                  | 62.70        | 28.90  | 27.23        | 25.15        | 35.90        |

Table 14: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies  $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$  and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

|                                                                                     | Clean        | AA           | NRR          |
|-------------------------------------------------------------------------------------|--------------|--------------|--------------|
| <i>(a) With the stack: WA, AWP, <math>\epsilon_{\text{train}} = 8.8/255</math></i>  |              |              |              |
| Raw features (squared $\ell_2$ )                                                    | 60.21        | <b>28.14</b> | 38.35        |
| Both normalized (cosine)                                                            | <b>61.41</b> | 28.07        | <b>38.53</b> |
| <i>(b) Base regime: no WA, no AWP, <math>\epsilon_{\text{train}} = 8/255</math></i> |              |              |              |
| Raw features (squared $\ell_2$ )                                                    | <b>62.40</b> | <b>24.34</b> | <b>35.02</b> |
| Both normalized (cosine)                                                            | 61.52        | 22.90        | 33.38        |
| Asymmetric: student normalized, teacher raw                                         | 61.28        | 22.98        | 33.43        |

Table 15:  $\epsilon_{\text{train}}$  swept with everything else fixed. Evaluation is at  $\epsilon = 8/255$  in every cell.

|                      | Clean        | PGD-20       | CW           | AA           | NRR          |
|----------------------|--------------|--------------|--------------|--------------|--------------|
| <i>CIFAR-100</i>     |              |              |              |              |              |
| 6/255                | <b>68.07</b> | 28.40        | 28.16        | 25.82        | 37.44        |
| 7/255                | 65.61        | 30.34        | 29.64        | 27.44        | 38.70        |
| 8/255                | 63.89        | 31.41        | 29.83        | 27.78        | 38.72        |
| 8.8/255 <sup>†</sup> | 62.17        | 32.37        | <b>30.93</b> | <b>28.86</b> | <b>39.42</b> |
| 10/255 <sup>†</sup>  | 59.99        | <b>32.82</b> | 30.53        | 28.77        | 38.89        |
| <i>CIFAR-10</i>      |              |              |              |              |              |
| 6/255                | <b>89.64</b> | 49.29        | 50.20        | 47.88        | 62.42        |
| 7/255                | 88.30        | 51.43        | 52.64        | 50.27        | 64.07        |
| 8/255 <sup>†</sup>   | 87.22        | 52.43        | 53.55        | 51.15        | <b>64.48</b> |
| 8.8/255 <sup>†</sup> | 84.96        | 52.98        | <b>53.94</b> | 51.74        | 64.31        |
| 10/255 <sup>†</sup>  | 83.29        | <b>53.25</b> | 53.88        | <b>51.79</b> | 63.87        |

Table 16: Effect of AWP on top of weight averaging across datasets and training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses  $\gamma = 0.005$  after a 10-epoch warmup. Within each row, AWP is the only added component.

|                  | WA    |       |       | WA + AWP |       |       |
|------------------|-------|-------|-------|----------|-------|-------|
|                  | Clean | AA    | NRR   | Clean    | AA    | NRR   |
| <i>CIFAR-100</i> |       |       |       |          |       |       |
| 8/255            | 63.54 | 26.75 | 37.65 | 63.89    | 27.78 | 38.72 |
| 8.8/255          | 62.30 | 27.69 | 38.34 | 62.17    | 28.86 | 39.42 |
| <i>CIFAR-10</i>  |       |       |       |          |       |       |
| 8/255            | 86.67 | 49.20 | 62.77 | 87.22    | 51.15 | 64.48 |
| 8.8/255          | 85.16 | 50.15 | 63.13 | 84.96    | 51.74 | 64.31 |

Table 17: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then without weight averaging or AWP.

|                                | CIFAR-100 |              |       | CIFAR-10 |              |       |
|--------------------------------|-----------|--------------|-------|----------|--------------|-------|
|                                | Clean     | AA           | NRR   | Clean    | AA           | NRR   |
| ARD                            | 57.61     | 20.24        | 29.96 | 84.58    | 43.75        | 57.67 |
| + teacher initialization       | 57.96     | 20.22        | 29.98 | —        | —            | —     |
| RSLAD                          | 59.68     | 21.30        | 31.40 | 85.15    | 42.91        | 57.06 |
| + teacher initialization       | 59.87     | 20.71        | 30.77 | —        | —            | —     |
| AdaAD                          | 59.79     | 23.19        | 33.42 | 85.39    | 46.22        | 59.98 |
| + teacher initialization       | 59.75     | 22.78        | 32.98 | —        | —            | —     |
| AdaAD + IGDM                   | 48.25     | 19.48        | 27.75 | 85.01    | 45.50        | 59.27 |
| + teacher initialization       | 49.67     | 20.15        | 28.67 | —        | —            | —     |
| PGD-AT, teacher initialization | 57.73     | <b>26.46</b> | 36.29 | 82.23    | <b>50.72</b> | 62.74 |
| CFA, no WA/AWP                 | 61.29     | 25.36        | 35.88 | —        | —            | —     |

Table 18: RPAT++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at  $\gamma = 0.005$  with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

|                                | Final |              |              | Best by PGD-20 |              |              |
|--------------------------------|-------|--------------|--------------|----------------|--------------|--------------|
|                                | Clean | AA           | NRR          | Clean          | AA           | NRR          |
| RPAT++ (its own recipe)        | 57.05 | 26.61        | 36.30        | 55.93          | 27.37        | 36.74        |
| + AWP                          | 57.48 | <b>28.05</b> | <b>37.70</b> | 57.50          | <b>27.92</b> | <b>37.60</b> |
| + AWP + teacher initialization | 57.83 | 27.47        | 37.25        | 57.80          | 27.59        | 37.36        |

## F THE ALLOCATION RULE AND ITS BUDGET SOLVE

### F.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an  $\ell_\infty$  ball of radius  $e$  is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (12)$$

the dual norm of  $\ell_\infty$  being  $\ell_1$ . Equalizing that movement across a batch subject to a fixed total budget  $\sum_i \epsilon_i = N\epsilon$  gives  $\epsilon_i g_i = \text{const}$ , that is  $\epsilon_i \propto 1/g_i$  with  $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$ . The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking  $\epsilon_i \propto (\bar{g}/g_i)^p$  makes  $p = 1$  that allocation,  $p = 0$  the uniform-pixel one, and  $p \in (0, 1)$  an interpolation.

Two approximations are worth declaring rather than leaving to be found. First,  $g_i$  is measured once, by a single backward pass at the *clean*  $x_i$  before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in  $\epsilon$ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective.  $\|\nabla_x \mathcal{L}\|_1$  is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- $\ell_2$  anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched*  $\epsilon$  rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the reported configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 5 is a comparison of which geometry the budget follows.

The first-order derivation and the reported implementation use different sensitivity norms. Equation (12) gives the  $\ell_1$  norm and the reported implementation uses the  $\ell_2$  gradient norm, which was fixed before this comparison and retained for reproducibility. The two are not the same allocation. Measured over training on the reported configuration, the  $\ell_1$  signal’s coefficient of variation has median 0.68 against the  $\ell_2$  signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the reported recipe with `featdir_angepts_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

The difference is small but non-negligible relative to the observed seed variation. It is twice the seed-to-seed spread of Appendix G, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius,  $p = 1$  gains +1.75 clean and +0.44 AutoAttack at  $\ell_2$  and +1.74 and +0.15 at  $\ell_1$ . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. Thus the sensitivity norm is a genuine implementation choice: the reported  $\ell_2$  variant performs slightly better in AA, while the clean-accuracy gain persists with the theoretically matched  $\ell_1$  norm.

The permuted control in Table 5 isolates the assignment signal from the radius distribution. It takes the multiset of weights the gradient rule produces — the same values, the same clip  $[0.5, 1.5]$ , the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

## F.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at  $10^{-12}$  and computes  $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$ , followed by  $w_i = Nv_i / \sum_j v_j$  and  $\epsilon_i = \epsilon w_i$ . PGD uses step size  $a_i = aw_i$ . This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all  $g_i$  equal and the allocation uniform.

**Optional exact-box variant.** The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With  $r_i = (\bar{g}/g_i)^p$  and box  $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$ , the radii that satisfy both the box and the budget exactly are  $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$  with  $t$  the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (13)$$

The left side is continuous and nondecreasing in  $t$  from  $Nw_{\text{lo}}$  to  $Nw_{\text{hi}}$ , so a root exists and bisection finds it; when no clip is active it reduces to  $t = N/\sum_i r_i$ , that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

Table 19: WideResNet-34-10 on CIFAR-100 at  $\epsilon = 8/255$ , measured in our framework.

| Method            | T.   | Clean        | AA           | Avg          | NRR          |
|-------------------|------|--------------|--------------|--------------|--------------|
| PGD-AT            | –    | 58.97        | 22.84        | 40.91        | 32.93        |
| TRADES            | –    | 57.35        | 25.60        | 41.48        | 35.40        |
| MART              | –    | 57.16        | 22.43        | 39.80        | 32.22        |
| ARD               | nat. | 59.52        | 23.05        | 41.29        | 33.25        |
| RSLAD             | nat. | 60.48        | 21.63        | 41.06        | 31.87        |
| AdaAD             | nat. | 61.66        | 25.62        | 43.64        | 36.20        |
| LBGAT             | nat. | 59.63        | 28.55        | 44.09        | 38.61        |
| ARREST            | nat. | <b>67.69</b> | 24.69        | 46.19        | 36.18        |
| HAT               | nat. | 62.81        | 26.91        | 44.86        | 37.68        |
| ADR               | self | 64.50        | 29.04        | 46.77        | 40.05        |
| <b>CFA (ours)</b> | nat. | 65.04        | <b>30.14</b> | <b>47.59</b> | <b>41.19</b> |

## G EXPERIMENTAL DETAILS

Table 20: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

| Dataset       | Architecture | Epochs | Clean |
|---------------|--------------|--------|-------|
| CIFAR-10      | ResNet-18    | 200    | 95.32 |
| CIFAR-100     | ResNet-18    | 200    | 77.66 |
| Tiny-ImageNet | ResNet-18    | 200    | 66.29 |

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 27: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

**Teacher.** One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 20. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; ?? is the ablation over how long it is trained.

**Student.** The main ResNet-18 configurations are listed below. Unless noted otherwise, CIFAR-100 ablations use 100 epochs at  $8.8/255$ , with a 10-epoch AWP warmup.

|                                             | CIFAR-10           | CIFAR-100          | Tiny-ImageNet-200  |
|---------------------------------------------|--------------------|--------------------|--------------------|
| Backbone                                    | ResNet-18          | ResNet-18          | ResNet-18          |
| Optimizer                                   | AdamW              | AdamW              | AdamW              |
| Learning rate / schedule                    | 0.021, one-cycle   | 0.021, one-cycle   | 0.021, one-cycle   |
| Epochs                                      | 150                | 150                | 100                |
| Batch size                                  | 128                | 128                | 128                |
| Training radius $\epsilon_{\text{train}}$   | 8/255              | 8/255              | 8.8/255            |
| Inner attack                                | PGD-10, step 2/255 | PGD-10, step 2/255 | PGD-10, step 2/255 |
| Allocation exponent $p$                     | 1                  | 1                  | 1                  |
| Radius box $[w_{\text{lo}}, w_{\text{hi}}]$ | [0.5, 1.5]         | [0.5, 1.5]         | [0.5, 1.5]         |
| Classifier head                             | frozen             | frozen             | frozen             |
| Weight averaging (start, decay)             | 0.2, 0.999         | 0.2, 0.999         | 0.2, 0.999         |
| AWP ( $\gamma$ , warmup)                    | 0.005, 15 ep       | 0.005, 15 ep       | 0.005, 10 ep       |
| Teacher checkpoint                          | clean, 200 ep      | clean, 200 ep      | clean, 200 ep      |

The head being frozen is what removes  $\tau$  and  $\beta$  from the method: with the classifier inherited and never updated, the head-KD term of earlier drafts has no parameters to reach and is switched off entirely.  $p$  is not tuned. The training-radius sweep is reported in Table 15; evaluation is at 8/255 in every cell.

**Evaluation.** We report AutoAttack accuracy at  $\ell_\infty$  radius 8/255 on the full test set; NRR is its harmonic mean with clean accuracy. Additional attacks are in Appendices J and L. Ported baselines retain their published recipes unless stated otherwise.

**Repeated training.** We repeat both arms of the final radius-allocation comparison with three student seeds. Uniform radii give  $62.63 \pm 0.22$  clean and  $28.01 \pm 0.20$  AA; sensitivity matching gives  $64.82 \pm 0.07$  clean and  $28.04 \pm 0.09$  AA (mean  $\pm$  sample standard deviation). The paired clean gains are 2.35, 2.24, and 1.97 points, whereas the paired AA changes are 0.14,  $-0.23$ , and 0.18. Thus the clean gain repeats, while the data support no AA change. Other rows remain individual runs, so small differences between regression targets or nearby operating points should not be interpreted as significant.

**Compute, teacher included.** A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of ?? included. Table 21 puts the total against the baselines measured on the same machine.

Table 21: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

| Method                             | Epochs    | Teacher (h) | Student (h) | Total (h)   |
|------------------------------------|-----------|-------------|-------------|-------------|
| Natural teacher (ours, reused)     | 200 clean | –           | 0.39        | 0.39        |
| <b>CFA (ours)</b>                  | 100 adv.  | 0.39        | 2.07        | <b>2.46</b> |
| Label CE at our regime             | 100 adv.  | 0.39        | 3.22        | 3.61        |
| PGD-AT, initialized at the teacher | 100 adv.  | 0.39        | 1.72        | 2.11        |
| ARD                                | 100 adv.  | 0.39        | 1.52        | 1.91        |
| RSLAD                              | 100 adv.  | 0.39        | 1.71        | 2.10        |
| AdaAD                              | 100 adv.  | 0.39        | 2.79        | 3.18        |
| LBGAT                              | 100 adv.  | joint       | 1.88        | 1.88        |
| HAT                                | 50 adv.   | 0.39        | 1.10        | 1.49        |
| ADR                                | 200 adv.  | self        | 3.52        | 3.52        |

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. And

the comparison with adversarial distillation is favourable in a way the table understates, because those rows assume a robust teacher that the current literature obtains from a WideResNet trained on generated data, which is more expensive than every row here combined.

## H ADDITIONAL TEACHER-CHECKPOINT RESULTS

?? shows one natural-training trajectory; Table 22 reports an independently trained trajectory. Each listed teacher was trained to completion under its own one-cycle schedule, then used to initialize a 50-epoch student with the same training procedure. These experiments vary the teacher checkpoint, not just the target features.

Table 22: Independent CIFAR-100 teacher trajectory (teacher seed 1; 50-epoch students with seed 0). Teacher metrics are measured on held-out data.

| Epoch | Teacher |        |             | Student |              |              |
|-------|---------|--------|-------------|---------|--------------|--------------|
|       | Clean   | Margin | Sensitivity | Clean   | AA           | NRR          |
| 20    | 73.87   | 3.074  | 3.911       | 63.82   | 23.22        | 34.05        |
| 40    | 75.57   | 3.931  | 4.542       | 64.36   | 23.86        | 34.81        |
| 100   | 77.27   | 4.506  | 4.678       | 63.68   | 25.61        | 36.53        |
| 150   | 77.68   | 4.495  | 4.262       | 63.37   | 25.49        | 36.36        |
| 200   | 77.90   | 4.542  | 4.129       | 63.02   | <b>26.16</b> | <b>36.97</b> |
| 300   | 78.27   | 4.523  | 3.835       | 62.34   | 25.94        | 36.64        |

The second trajectory again moves from higher student clean accuracy at earlier checkpoints to higher AA at later ones. The strongest AA occurs at 200 rather than 300 epochs, although the difference is small. Feature sensitivity has only moderate rank correlation with student clean accuracy on these six checkpoints (Spearman 0.54, versus 0.83 on the same grid of the first trajectory). The matched teachers in ?? further show that teacher clean accuracy alone does not determine student clean accuracy. None of these measurements supplies a validated teacher-only selection rule.

## I THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines retain their authors’ recipes. Table 23 asks whether that recipe is what wins: it runs our own objective, and label cross-entropy for reference, under both recipes and both initializations. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 23: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, the schedule PGD-AT uses here. The training radius and every other field are held fixed within each pair.

| Objective           | Recipe               | Clean        | AA           | Avg          | NRR          |
|---------------------|----------------------|--------------|--------------|--------------|--------------|
| Anchor              | ours, teacher init   | <b>61.29</b> | 25.36        | <b>43.33</b> | <b>35.88</b> |
| Anchor              | ours, random init    | 60.81        | 25.21        | 43.01        | 35.66        |
| Anchor              | theirs, teacher init | 59.65        | <b>25.54</b> | 42.60        | 35.77        |
| Label cross-entropy | ours, teacher init   | 54.60        | 19.45        | 37.03        | 28.68        |
| Label cross-entropy | theirs, teacher init | 56.66        | 20.91        | 38.79        | 30.55        |

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, so Table 3 includes that stronger baseline. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

Table 24: Successive additions to the CIFAR-100 ResNet-18 anchor (100 epochs,  $\epsilon_{\text{train}} = 8.8/255$ ).

|                                  | Clean        | AA           | NRR          |
|----------------------------------|--------------|--------------|--------------|
| Anchor ( $p = 0$ )               | 61.29        | 25.36        | 35.88        |
| + sensitivity-matched $\epsilon$ | <b>63.09</b> | 25.74        | 36.56        |
| + weight averaging               | 62.30        | 27.69        | 38.34        |
| + AWP                            | 62.17        | <b>28.86</b> | <b>39.42</b> |

## J GRADIENT MASKING

We test for gradient masking with AutoAttack, independent attacks, and PGD sweeps (Athalye et al., 2018). Table 25 reports these checks for a 100-epoch CIFAR-100 repeat with 62.00% clean accuracy.

Table 25: Gradient-masking checks on all 10,000 CIFAR-100 test images at 8/255. Individual attacks are evaluated independently.

| Check                                            | Setting                         | Accuracy                      |
|--------------------------------------------------|---------------------------------|-------------------------------|
| <i>white box against black box</i>               |                                 |                               |
| APGD-CE                                          | white box, untargeted           | 31.80                         |
| APGD-T                                           | white box, targeted             | <b>28.83</b>                  |
| FAB-T                                            | white box, boundary             | 29.16                         |
| Square                                           | <b>black box</b> , 5000 queries | 34.82                         |
| AutoAttack                                       | standard evaluation             | 28.73                         |
| <i>radius sweep, PGD-20</i>                      |                                 |                               |
| $\epsilon = 2/255$                               |                                 | 54.62                         |
| $\epsilon = 4/255$                               |                                 | 47.03                         |
| $\epsilon = 8/255$                               |                                 | 32.46                         |
| $\epsilon = 12/255$                              |                                 | 20.88                         |
| $\epsilon = 16/255$                              |                                 | 11.94                         |
| $\epsilon = 32/255$                              |                                 | 1.05                          |
| $\epsilon = 64/255$                              |                                 | <b>0.00</b>                   |
| <i>step sweep, <math>\epsilon = 8/255</math></i> |                                 |                               |
| PGD-10 / 20 / 50 / 100                           |                                 | 33.17 / 32.57 / 32.30 / 32.24 |
| <i>transfer</i>                                  |                                 |                               |
| Natural teacher $\rightarrow$ student            | PGD-20 built on the teacher     | 60.42                         |

AutoAttack yields 28.73% accuracy. PGD accuracy falls monotonically with attack radius, reaching zero at 64/255, and decreases with additional steps. These checks show no indication of gradient masking under the evaluated attacks.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.4 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

## K NOTES ON THE BASELINES

Several rows of Table 1 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

**ARREST** uses its published objective: adversarial finetuning from the natural model with SGD at 0.025, the angular representation term at  $\lambda = 50$ , and Noisy Replay over the first half at  $\phi = 30^\circ$ . Table 1 reports its 150-epoch run for the main comparison, whereas Table 10 uses its published 20-epoch schedule. We read its representation after global pooling, where our anchor and geometry measurements are defined.

**LBGAT, and why its CIFAR-10 cell is empty.** The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and  $\text{lr} = 0.1$  is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and  $\beta = 6$  is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one  $\beta$  across both.

**HAT** uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its 61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

**ADR** is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

**What the published tables are blocked by.** Table 27 and Table 28 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is `ResNet(Bottleneck, [2, 2, 2, 2])` in its released code, 14.35M parameters against the 11.27M of the standard `BasicBlock` network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use `PreActResNet-18` explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is `PreActResNet-18`: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says `PreAct`, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F<sup>2</sup>AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

**IGDM** is reported at  $\alpha = 1$ . Its published  $\alpha = 20$  diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 17.

**Teacher-only KL with the training stack.** This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting,  $\tau = 4$ , gives 56.97/26.96 clean/AA;  $\tau = 16$  gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 11, where it remains competitive with feature regression.

## L PER-ATTACK METRICS

Table 26 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Table 26: Per-attack accuracy (%) for our cells. ResNet-18,  $\epsilon = 8/255$ , final-epoch weight-averaged model, full test set.

| Dataset   | Cell                                            | Clean | FGSM  | PGD-20 | CW    | AA    | NRR   |
|-----------|-------------------------------------------------|-------|-------|--------|-------|-------|-------|
| CIFAR-100 | CFA, head frozen (shipped)                      | 62.17 | 36.91 | 32.37  | 30.93 | 28.86 | 39.42 |
|           | the same cell at seed 1                         | 62.00 | 36.79 | 32.49  | 30.82 | 28.74 | 39.27 |
|           | $\ell_1$ gradient norm                          | 62.16 | 36.48 | 31.97  | 30.56 | 28.57 | 39.15 |
|           | CFA, head-KD term retained                      | 62.35 | 38.80 | 36.26  | 30.65 | 28.68 | 39.29 |
|           | CFA, $\epsilon_{\text{tr}} = 6/255$             | 68.07 | 35.88 | 28.40  | 28.16 | 25.82 | 37.44 |
|           | CFA, $\epsilon_{\text{tr}} = 7/255$             | 65.61 | 36.32 | 30.34  | 29.64 | 27.44 | 38.70 |
|           | CFA, $\epsilon_{\text{tr}} = 8/255$             | 63.89 | 36.67 | 31.41  | 29.83 | 27.78 | 38.72 |
|           | CFA, $\epsilon_{\text{tr}} = 10/255$            | 59.99 | 36.50 | 32.82  | 30.53 | 28.77 | 38.89 |
|           | CFA, no WA / AWP                                | 61.29 | 33.97 | 28.23  | 27.19 | 25.36 | 35.88 |
|           | Teacher-only KL, our stack                      | 56.97 | 35.82 | 33.68  | 28.94 | 26.96 | 36.60 |
|           | RPAT++ (our reproduction)                       | 55.93 | 34.46 | 32.44  | 29.37 | 27.36 | 36.74 |
|           | RPAT++ (our reproduction)                       | 55.93 | 34.46 | 32.44  | 29.37 | 27.36 | 36.74 |
| CIFAR-10  | CFA, $\epsilon_{\text{tr}} = 8.8/255$ (shipped) | 84.96 | 59.58 | 52.98  | 53.94 | 51.74 | 64.31 |
|           | CFA, $\epsilon_{\text{tr}} = 6/255$             | 89.64 | 59.37 | 49.29  | 50.20 | 47.88 | 62.42 |
|           | CFA, $\epsilon_{\text{tr}} = 7/255$             | 88.30 | 59.59 | 51.43  | 52.64 | 50.27 | 64.07 |
|           | CFA, $\epsilon_{\text{tr}} = 8/255$             | 87.22 | 59.92 | 52.43  | 53.55 | 51.15 | 64.48 |
|           | CFA, $\epsilon_{\text{tr}} = 10/255$            | 83.29 | 58.98 | 53.25  | 53.88 | 51.79 | 63.87 |
|           | RPAT++ (our reproduction)                       | 82.41 | 59.73 | 55.72  | 52.88 | 50.75 | 62.82 |

**Why PGD-20 is not the metric here.** The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and  $\text{CW}_\infty$ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 26 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

|                       | PGD-20 | CW    | AA    |
|-----------------------|--------|-------|-------|
| head-KD term retained | 36.26  | 30.65 | 28.68 |
| head frozen           | 32.63  | 30.66 | 28.77 |
| difference            | −3.63  | +0.01 | +0.09 |

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

## M PUBLISHED RESULTS, FOR CONTEXT

Table 1 contains only numbers we measured ourselves. Table 27 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Read against Table 1 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 64.82/28.04 at Avg 46.43 on CIFAR-100 and 87.37/51.33 at Avg 69.35 on CIFAR-10.

Table 27: Published results,  $\ell_\infty$ ,  $\epsilon = 8/255$ , quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 1. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

| Method                            | Teacher | Clean | AA    | Avg   | NRR   | Source            |
|-----------------------------------|---------|-------|-------|-------|-------|-------------------|
| <i>CIFAR-100, ResNet-18</i>       |         |       |       |       |       |                   |
| PGD-AT                            | –       | 56.56 | 25.02 | 40.79 | 34.69 | RPAT              |
| TRADES                            | –       | 55.39 | 24.51 | 39.95 | 33.98 | RPAT              |
| MART                              | –       | 49.83 | 25.00 | 37.41 | 33.30 | RPAT              |
| Consistency-AT                    | –       | 58.53 | 25.39 | 41.96 | 35.42 | RPAT              |
| Consistency-AT + RPAT             | –       | 60.33 | 26.31 | 43.32 | 36.64 | RPAT              |
| F <sup>2</sup> AT                 | –       | 54.19 | 23.24 | 38.71 | 32.53 | F <sup>2</sup> AT |
| Generalist++                      | co-tr.  | 62.97 | 23.96 | 43.47 | 34.71 | Generalist++      |
| ADR (AT + ADR)                    | self    | 56.10 | 26.87 | 41.48 | 36.34 | ADR               |
| ADR (AT + WA + AWP + ADR)         | self    | 57.36 | 28.50 | 42.93 | 38.08 | ADR               |
| <i>CIFAR-10, ResNet-18</i>        |         |       |       |       |       |                   |
| PGD-AT                            | –       | 82.78 | 44.63 | 63.71 | 57.99 | CURE              |
| TRADES                            | –       | 82.41 | 48.37 | 65.39 | 60.96 | CURE              |
| MART                              | –       | 80.70 | 47.49 | 64.10 | 59.79 | CURE              |
| ST-AT                             | –       | 83.10 | 50.50 | 66.80 | 62.82 | CURE              |
| IAD                               | robust  | 80.63 | 50.17 | 65.40 | 61.85 | CURE              |
| Consistency-AT + RPAT             | –       | 84.12 | 48.98 | 66.55 | 61.91 | RPAT              |
| Generalist++                      | co-tr.  | 89.09 | 46.07 | 67.58 | 60.73 | Generalist++      |
| ARREST                            | natural | 86.63 | 46.14 | 66.39 | 60.21 | ARREST            |
| CURE                              | self    | 86.76 | 49.69 | 68.23 | 63.19 | CURE              |
| ADR (AT + ADR)                    | self    | 82.41 | 50.38 | 66.40 | 62.53 | ADR               |
| <i>CIFAR-10, PreActResNet-18</i>  |         |       |       |       |       |                   |
| WA (Izmailov et al., 2018)        | –       | 83.50 | 49.89 | 66.70 | 62.46 | RPAT              |
| AWP (Wu et al., 2020)             | –       | 81.11 | 50.09 | 65.60 | 61.93 | RPAT              |
| KD + SWA (Chen et al., 2021)      | robust  | 84.06 | 49.82 | 66.94 | 62.56 | RPAT              |
| TE (Dong et al., 2022)            | –       | 82.04 | 50.12 | 66.08 | 62.23 | RPAT              |
| ReBAT (Wang et al., 2023)         | –       | 82.09 | 50.72 | 66.41 | 62.70 | RPAT              |
| RPAT++ (Wang & Liu, 2025)         | –       | 82.63 | 51.00 | 66.82 | 63.07 | RPAT              |
| <i>CIFAR-100, PreActResNet-18</i> |         |       |       |       |       |                   |
| WA (Izmailov et al., 2018)        | –       | 57.26 | 25.83 | 41.55 | 35.60 | RPAT              |
| AWP (Wu et al., 2020)             | –       | 54.10 | 25.16 | 39.63 | 34.35 | RPAT              |
| KD + SWA (Chen et al., 2021)      | robust  | 57.17 | 25.66 | 41.42 | 35.42 | RPAT              |
| TE (Dong et al., 2022)            | –       | 56.41 | 25.84 | 41.13 | 35.44 | RPAT              |
| ReBAT (Wang et al., 2023)         | –       | 56.13 | 27.60 | 41.87 | 37.00 | RPAT              |
| RPAT++ (Wang & Liu, 2025)         | –       | 56.84 | 27.68 | 42.26 | 37.23 | RPAT              |
| <i>CIFAR-10, WideResNet-34-10</i> |         |       |       |       |       |                   |
| WA (Izmailov et al., 2018)        | –       | 87.66 | 52.65 | 70.16 | 65.79 | RPAT              |
| MMA (Ding et al., 2020)           | –       | 87.80 | 43.10 | 65.45 | 57.82 | RPAT              |
| AWP (Wu et al., 2020)             | –       | 85.63 | 53.32 | 69.48 | 65.72 | RPAT              |
| KD + SWA (Chen et al., 2021)      | robust  | 87.45 | 53.59 | 70.52 | 66.46 | RPAT              |
| TE (Dong et al., 2022)            | –       | 85.97 | 52.88 | 69.43 | 65.48 | RPAT              |
| ReBAT (Wang et al., 2023)         | –       | 85.25 | 54.78 | 70.02 | 66.70 | RPAT              |
| ADR (Wu et al., 2024)             | self    | 84.67 | 53.25 | 68.96 | 65.38 | RPAT              |
| CURE (Gowda et al., 2024)         | self    | 87.05 | 52.10 | 69.58 | 65.19 | RPAT              |
| RPAT++ (Wang & Liu, 2025)         | –       | 86.76 | 54.97 | 70.87 | 67.30 | RPAT              |

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Tables 2 and 19 are where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 1 is for.

Table 28: Published Tiny-ImageNet-200 results, ResNet-18,  $\ell_\infty$ ,  $\epsilon = 8/255$ , quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix K says how each differs.

| Method                                                                   | Clean        | AA           | Avg          | NRR          | Source            |
|--------------------------------------------------------------------------|--------------|--------------|--------------|--------------|-------------------|
| <b>CFA (ours)</b>                                                        | <b>55.16</b> | <b>20.54</b> | <b>37.85</b> | <b>29.93</b> | —                 |
| AT + WA + ADR                                                            | 48.55        | 20.23        | 34.39        | 28.56        | ADR               |
| AT + WA + AWP + ADR                                                      | 48.27        | 20.12        | 34.20        | 28.40        | ADR               |
| TRADES + WA + AWP + ADR                                                  | 51.38        | 19.48        | 35.43        | 28.25        | ADR               |
| TRADES + WA + ADR                                                        | 51.99        | 19.17        | 35.58        | 28.01        | ADR               |
| AT + WA + AWP                                                            | 48.61        | 19.58        | 34.09        | 27.92        | ADR               |
| AT + WA                                                                  | 49.10        | 19.30        | 34.20        | 27.71        | ADR               |
| TE*                                                                      | 47.46        | 18.29        | 32.88        | 26.40        | ADR               |
| AT                                                                       | 45.87        | 18.06        | 31.96        | 25.92        | ADR               |
| TRADES                                                                   | 48.49        | 17.35        | 32.92        | 25.56        | ADR               |
| <i>PreActResNet-18 — same parameter count, different network</i>         |              |              |              |              |                   |
| HAT                                                                      | 52.60        | 18.14        | 35.37        | 26.98        | HAT               |
| TRADES                                                                   | 48.25        | 17.17        | 32.71        | 25.35        | HAT               |
| AT                                                                       | 47.76        | 17.92        | 32.84        | 26.02        | HAT               |
| <i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i> |              |              |              |              |                   |
| Consistency-AT + RPAT                                                    | 49.74        | 18.84        | 34.29        | 27.33        | RPAT              |
| PGD-AT + RPAT                                                            | 47.68        | 17.77        | 32.73        | 25.89        | RPAT              |
| Consistency-AT                                                           | 46.54        | 17.60        | 32.07        | 25.54        | RPAT              |
| MART                                                                     | 39.70        | 17.18        | 28.44        | 23.98        | RPAT              |
| F <sup>2</sup> AT                                                        | 40.54        | 13.13        | 26.84        | 19.84        | F <sup>2</sup> AT |
| SEAT                                                                     | 35.51        | 11.37        | 23.44        | 17.22        | F <sup>2</sup> AT |
| SAT                                                                      | 31.52        | 9.37         | 20.45        | 14.45        | F <sup>2</sup> AT |

## N THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74  $\rightarrow$  45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 27 is therefore the paper’s own number, and CURE is omitted from Table 1.

## O A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the

AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen after the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the reported recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the reported recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix L and the  $\epsilon_{\text{train}}$  sweep in Table 15 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

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